A FIELD GUIDE TO THE ALIEN ARTIFACT

STARCROSS

Maximum-score walkthrough

A deterministic, command-by-command route to Earth and all 400 points

VERIFIED FINISH

400 / 400

Earth course • 252 moves • 226 commands • no deaths

188,557 executed instructions • 57 unique opcode rows

How to play this route

Start a fresh Starcross Release 17 game. Enter one numbered command at each prompt. On each route page read down the left column, then down the right. The repeated WAIT and AGAIN entries are intentional.

The mass detector selects UM28. Set the computer to R 250, THETA 45, PHI 178, confirm, sit, and fasten the belt. Keep the safety line and space suit for the exterior rods.

The route uses the machine's 400-point total as its score ceiling. Its final blue-button action selects an Earth encounter on an ellipse and ends with the artifact following the chosen course.

Score checkpoints

25 pts Red Dock	**75 pts** Nesting Cage	**150 pts** Observatory	**225 pts** Laboratory
250 pts Village Center	**300 pts** Garage	**350 pts** Control Bubble	**400 pts** Earth course

Replay settings: deterministic seed 1035 • mass UM28 • clean Release 17 game • route and trace stay local.

The maps index all 86 room objects (59 short-name labels). Red arrows mark selected transitions verified on this route.

Opcode inventory is derived from the same fresh replay; "First Seq" is one-based and "First PC" is hexadecimal.

Commands 001–076

From the bunk to the first alien halls. The route enters the artifact at the Red Dock and finishes this section with the gold rod.

Checkpoint: Red Dock 25 • Nesting Cage 75 • Spider ship 100

WAKE AND SET THE COURSE — UM28: R 250 / THETA 45 / PHI 178

001. get up
CHECKPOINT 0 / 400 • Living Quarters
002. get library
003. east
004. press red button
005. read screen
006. computer, r is 250. theta is 45. phi is 178
007. computer, confirm
008. sit
009. fasten belt
010. wait
011. wait
012. wait
013. wait
014. wait
015. wait
016. wait
017. wait
018. wait
019. wait

BOARD THE ARTIFACT — suit up, solve the solar-system sculpture, enter Red Hall

020. unfasten belt
021. get up
022. east
023. get suit
024. get safety line
025. west
026. open inner door
027. out
028. close inner door
029. wear suit
030. open outer door
031. out
CHECKPOINT 25 / 400 • Red Dock
032. examine sculpture
033. press fourth bump
034. push tiny column
035. get black rod
CHECKPOINT 50 / 400 • Red Dock
036. in
037. close outer door
038. remove suit
039. open inner door
040. up

RED AND YELLOW RODS — clear the zoo nest, then bargain with the spider

041. north
042. west
043. north
044. east
045. throw library at nest
046. get library
047. get red rod
CHECKPOINT 75 / 400 • Nesting Cage
048. west
049. south
050. west
051. south
052. open inner door
053. down
054. close inner door
055. open outer door
056. down
057. aft
058. aft
059. give library
060. get yellow rod
CHECKPOINT 100 / 400 • Spherical Ship
061. fore
062. fore
063. up
064. close outer door
065. open inner door
066. up
067. up

REPAIR THE ARTIFACT — lights, air circulation, and the computer

068. south
069. open metal hatch
070. down
071. take square
072. put yellow rod in yellow slot
073. put red rod in second red slot
074. up
075. north
076. down

Commands 077–152

Recover clear, silver, and pink; prepare the disk machine; then trade the suit for the brown rod and pursue the chief.

Checkpoint: Observatory 150 • Pink rod 200 • Laboratory 225

077. east
078. north
079. east
080. north
081. east
082. south
083. open panel
084. put square in slot
085. close panel
086. turn computer on
087. get gold rod
CHECKPOINT 125 / 400 • Computer Room

CLEAR, SILVER, AND PINK RODS — observatory, ray gun, basket, and tether

088. north
089. east
090. east
091. south
092. west
093. look in projector through black rod
094. get clear rod
CHECKPOINT 150 / 400 • Observatory
095. drop black rod
096. east
097. south
098. south
099. south
100. west
101. take ray gun
102. look in barrel
CHECKPOINT 175 / 400 • Weapons Deck
103. get silver rod
104. east
105. north
106. west
107. open inner door
108. down
109. get basket
110. put all rods in basket
111. close inner door
112. wear suit
113. open outer door
114. again
115. down
116. tie line to suit
117. tie line to hook
118. west
119. get pink rod
CHECKPOINT 200 / 400 • Among Debris

THE LABORATORY — generate the blue rod; leave the red disk for the mouse

120. east
121. untie line from hook
122. up
123. close outer door
124. remove suit
125. drop line
126. open inner door
127. up
128. south
129. south
130. east
131. get red disk
132. get blue disk
133. put red disk under globe
134. drop blue disk
135. put ray gun on globe
136. turn dial to 4
CHECKPOINT 225 / 400 • Laboratory
137. turn dial to 1
138. get all
139. put all rods in basket
140. west
141. south
142. west
143. drop red disk

THE WEASEL CHIEF — trade the suit for the brown rod and follow him

144. west
145. north
146. north
147. east
148. east
149. wait
150. wait
151. wait
152. wait

Commands 153–226

Follow the chief to the Green Airlock, teleport to the Garage, collect the final rods, and fly the artifact to Earth.

Checkpoint: Brown rod 250 • Garage 300 • Earth course 400

153. wait
154. give suit
155. point at brown rod
CHECKPOINT 250 / 400 • Village Center
156. follow chief
157. again
158. again
159. again
160. again
161. again
162. again
163. again
164. again
165. again
166. again
167. again
168. again

GREEN AIRLOCK AND VIOLET ROD — descend from the Warren center

169. down
170. close inner door
171. open outer door
172. down
173. west
174. west
175. fore
176. touch skeleton
177. look
178. get violet rod
CHECKPOINT 275 / 400 • Control Room

GARAGE TELEPORT — drop the blue disk, then search the bin for green

179. drop blue disk
180. stand on disk
181. search bin
182. again
183. again
184. get green rod
CHECKPOINT 300 / 400 • Garage
185. put all rods in basket

THE DRIVE BUBBLE — climb to the treetop and restore the star drive

186. north
187. west
188. north
189. north
190. up
191. east
192. south
193. south
194. climb tree
195. up
196. jump
197. put silver rod in silver slot
198. in
199. get white rod
CHECKPOINT 325 / 400 • Drive Bubble
200. put white rod in white slot
201. out
202. up
203. jump
204. shoot drive bubble
205. again
206. again
207. down
208. put gold rod in gold slot

CONTROL BUBBLE — insert the gold and six colored rods

209. in
CHECKPOINT 350 / 400 • Control Bubble
210. put clear rod in clear slot
211. put brown rod in brown slot
212. put green rod in green slot
213. put blue rod in blue slot
214. put violet rod in violet slot
215. put pink rod in pink slot
216. press large square

SET EARTH — select Earth, choose an ellipse, raise speed, press blue

217. press brown spot
218. again
219. again
220. again
221. press violet spot
222. again
223. again
224. press green spot
225. again
226. press blue button
CHECKPOINT 400 / 400 • Control Bubble

Ship And Red Entry

Room IDs are story object numbers; repeated names group rooms with the same printed label.

Living Quarters
ID 10

E

Bridge
ID 202

E

Storage
ID 198

Airlock
ID 122

Outside Ship
ID 197

Red Dock
ID 188

IN

Red Airlock
ID 175

UP

Red Hall
ID 20, 92, 154, 167, 186

Room on Ring Two
ID 19

N

Zoo
ID 46

E

Nesting Cage
ID 102

Blue Hall
ID 53, 68, 105, 118, 166, 194

Blue Airlock
ID 106

Control Room
ID 15

Fore End
ID 99, 190

Artifact Ring And Workrooms

Room IDs are story object numbers; repeated names group rooms with the same printed label.

Green Hall
ID 32, 67, 234

Yellow Hall
ID 52, 91, 117, 183, 184, 185

Room on Ring One
ID 115

Room on Ring Four
ID 211

E

S

Repair Room
ID 148

Computer Room
ID 137

Observatory
ID 153

Weapons Deck
ID 147

Laboratory
ID 80

Melted Spot
ID 90

Metal Floor
ID 134, 224

Guard Room
ID 149

Broken Cage
ID 208

Ports, Alien Ships, And Village

Room IDs are story object numbers; repeated names group rooms with the same printed label.

Blue Dock
ID 94

Green Airlock
ID 69

DOWN

Green Dock
ID 33

Yellow Airlock
ID 142

W

Yellow Dock
ID 187

Umbilical
ID 64

W

Cargo Hold
ID 54

Spherical Ship
ID 35

W

Bubbles
ID 55

Among Debris
ID 143

Outskirts of Village
ID 17, 50, 51, 81, 172, 193

Village Suburbs
ID 215

Village Center
ID 141

FOLLOW

In the Warren
ID 216

AGAIN

Center of the Warren
ID 152

Forest, Drive, And Control Bubbles

Room IDs are story object numbers; repeated names group rooms with the same printed label.

Garage
ID 16

Grassland
ID 77, 225

S

Thin Forest
ID 43, 78

Dense Forest
ID 135

Base of Tree
ID 44

CLIMB

Up a Tree
ID 5

UP

Top of Tree
ID 159

JUMP

Drive Bubble Entrance
ID 136

Drive Bubble
ID 226

On Drive Bubble
ID 4

JUMP

Floating in Air
ID 27, 63, 112

SHOOT ×3

Floating in Space
ID 95

Control Bubble Entrance
ID 3

IN

Control Bubble
ID 223

On Control Bubble
ID 207

Inside Grue Cage
ID 230

Opcode table 1 of 3

Rows are ordered by first appearance in the successful fresh replay.

57 unique form/opcode/mnemonic rows **188,557 total instructions executed**

First Seq is one-based execution order; First PC is the instruction address in hexadecimal.

FORM	OPCODE	MNEMONIC	COUNT	FIRST SEQ	FIRST PC
Variable	0	call_vs	12,810	1	0x4391
Long	20	add	7,503	3	0x42FF
Long	1	je	20,542	5	0x4307
Long	21	sub	2,870	6	0x430B
Short	0	jz	31,964	7	0x430F
Variable	1	storew	5,582	9	0x431A
Short	11	ret	1,905	10	0x431F
Long	15	loadw	14,280	19	0x4321
Short	12	jump	13,588	22	0x432F
Long	13	store	14,257	32	0x43AB
Long	14	insert_obj	339	35	0x43B4
Long	10	test_attr	6,079	54	0x440B
Short	2	print	574	56	0x6085
Variable	9	and	1	58	0x611A
Variable	6	print_num	5	59	0x6120
Long	5	inc_chk	6,294	61	0x6132
Long	16	loadb	5,965	62	0x6139
Variable	5	print_char	6	63	0x613D
Short	11	new_line	550	87	0x6143
Short	0	rtrue	1,508	88	0x6144
Variable	8	push	1,546	95	0x5BA0
Variable	9	pull	206	97	0x5BA9

Opcode table 2 of 3

Rows are ordered by first appearance in the successful fresh replay.

57 unique form/opcode/mnemonic rows **188,557 total instructions executed**

First Seq is one-based execution order; First PC is the instruction address in hexadecimal.

FORM	OPCODE	MNEMONIC	COUNT	FIRST SEQ	FIRST PC
Long	11	set_attr	195	100	0x5BD8
Long	6	jin	1,477	102	0x5BDE
Short	10	print_obj	184	103	0x5BE2
Short	3	get_parent	2,264	106	0x5BEC
Long	17	get_prop	1,856	113	0x5C07
Short	13	print_paddr	143	119	0x5C1D
Short	2	get_child	2,263	131	0x5C68
Variable	1	je	5,397	144	0x5DA1
Short	1	get_sibling	2,250	149	0x5DF9
Short	5	inc	652	182	0x5E5B
Short	1	rfalse	1,425	212	0x7131
Short	8	ret_popped	1,481	252	0x5C84
Long	4	dec_chk	1,644	306	0x47D3
Variable	4	read	226	312	0x47E3
Long	7	test	3,170	328	0x4A59
Long	3	jg	3,073	333	0x4A5D
Long	9	and	1,035	334	0x4A61
Long	22	mul	553	344	0x491C
Variable	2	storeb	292	347	0x4928
Long	2	jl	2,436	397	0x49A2
Variable	7	random	309	756	0x7BE1
Variable	13	store	131	1,158	0x5500

Opcode table 3 of 3

Rows are ordered by first appearance in the successful fresh replay.

57 unique form/opcode/mnemonic rows **188,557 total instructions executed**

First Seq is one-based execution order; First PC is the instruction address in hexadecimal.

FORM	OPCODE	MNEMONIC	COUNT	FIRST SEQ	FIRST PC
Long	18	get_prop_addr	3,479	1,168	0x57F5
Short	4	get_prop_len	2,026	1,182	0x5ABE
Long	23	div	1,360	1,183	0x5AC1
Long	12	clear_attr	109	1,393	0x68C4
Short	3	print_ret	135	1,400	0x67BA
Variable	3	jg	18	4,252	0x4C6D
Short	7	print_addr	3	10,948	0x524E
Long	24	mod	13	14,333	0x9428
Short	9	remove_obj	129	15,902	0x9ACB
Long	19	get_next_prop	425	15,925	0x9D7E
Variable	3	put_prop	16	24,600	0x5F26
Short	6	dec	13	124,815	0xD408
Short	10	quit	1	188,557	0x6026

Inventory count check: 188,557 = 188,557 instructions.